Surjections from the positive integers and from the natural numbers transfer

The Open Logic Project contributors

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Editorial note. The counted statement is the author proposition labelled prop:enum-shift (879-882); the proof span 884-889 is the author proof environment that immediately follows it.

Editorial scope. One original result of the Open Logic Project unit sets-functions-relations/size-of-sets, counted once: a surjection from the positive integers onto A exists iff a surjection from the natural numbers onto A exists. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Surjection. A function f from X to A is surjective when every element of A is a value of f.. Both directions of the proof exhibit a function and check it is surjective by unpacking this definition (929-931, 932-933).
- Positive integers and natural numbers. The positive integers start at 1 and the natural numbers start at 0; adding 1 maps the natural numbers into the positive integers and subtracting 1 maps the positive integers into the natural numbers.. The shifts g(n) = f(n+1) and f(n) = g(n-1) move between the two domains (929, 932).

Proposition 0.1. *There is surjection $f\colon \mathbb{Z}^+ \to A$ iff there is surjection $g\colon \mathbb{N} \to A$.*

Proof. Given surjection $f\colon \mathbb{Z}^+ \to A$, we can define $g(n) = f(n+1)$ for all $n \in \mathbb{N}$. It is easy to see that $g\colon \mathbb{N} \to A$ is surjective. Conversely, given surjection $g\colon \mathbb{N} \to A$, define $f(n) = g(n-1)$. □